ENERG Y IJA IE IA

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Midea

MHA-V16W/D2N1
SMK-160/CD30GN1-B

	55°C	35
	A+	A++

A++
A+
A
B
C
D
E
F
G

45dB

72dB

	15	16
	15	**16**
	15	16
	kW	kW

2015

811/2013

Technical parameters

Model(s):	Outdoor unit: MHA-V16W/D2N1 Indoor unit: SMK-160/CD30GN1-B
Air-to-water heat pump:	YES
Water-to-water heat pump:	NO
Brine-to-water heat pump:	NO
Low-temperature heat pump:	NO
Equipped with a supplementary heater:	YES
Heat pump combination heater:	NO
Declared climate condition:	AVERAGE
Parameters are declared for medium-temperature application.	

Item	Symbol	Value	Unit
Rated heat output (*)	Prated	15	kW
Declared capacity for heating for part load at indoor temperature 20 °C and outdoor temperature Tj			
Tj = -7℃	Pdh	11.7	kW
Tj = 2℃	Pdh	8.1	kW
Tj = 7℃	Pdh	5.4	kW
Tj = 12℃	Pdh	2.8	kW
Tj = bivalent temperature	Pdh	12.1	kW
Tj = operating limit	Pdh	10.2	kW
For air-to-water heat pumps: Tj = -15℃	Pdh	-	kW
Bivalent temperature	T_{biv}	-5	°C
Cycling interval capacity for heating	P_{cych}	-	kW
Degradation co-efficient (**)	C_{dh}	0.9	--
Power consumption in modes other than active mode			
Off mode	P_{off}	0.019	kW
Standby mode	P_{sb}	0.019	kW
Thermostat-off mode	P_{to}	0.078	kW
Crankcase heater mode	P_{ck}	0.014	kW

Item	Symbol	Value	Unit
Seasonal space heating energy efficiency	ηs	127	%
Declared coefficient of performance or primary energy ratio for part load at indoor temperature 20 °C and outdoor temperature Tj			
Tj = -7℃	COPd	1.99	-
Tj = 2℃	COPd	3.09	-
Tj = 7℃	COPd	4.73	-
Tj = 12℃	COPd	6.81	-
Tj = bivalent temperature	COPd	2.15	-
Tj = operating limit	COPd	1.70	-
For air-to-water heat pumps: Tj = -15℃	COPd	-	-
For air-to-water heat pumps: Operation limit temperature	TOL	-10	°C
Cycling interval efficiency	COP_{cyc}	-	-
Heating water operating limit temperature	W_{TOL}	60	°C
Supplementary heater			
Rated heat output (**)	Psup	4.3	kW
Type of energy input	Electrical		

Other items			
Capacity control	variable		
Sound power level, indoors/ outdoors	L_{WA}	45/72	dB
Annual energy consumption	Q_{HE}	9491	kWh

For air-to-water heat pumps: Rated air flow rate, outdoors	–	6500	m^3/h
For water- or brine-to-water heat pumps: Rated brine or water flow rate, outdoor heat exchanger	–	-	m^3/h

For heat pump combination heater:							
Declared load profile	-			**Water heating energy efficiency**	η_{wh}	-	%
Daily electricity consumption	Q_{elec}	-	kWh	Daily fuel consumption	Q_{fuel}	-	kWh
Annual electricity consumption	AEC	-	kWh	Annual fuel consumption	AFC	-	GJ

Contact details	GD Midea Heating & Ventilating Equipment Co. Ltd（Penglai industry road, Beijiao, Shunde, Foshan, Guangdong, P.R China)

(*) For heat pump space heaters and heat pump combination heaters, the rated heat output Prated is equal to the design load for heating Pdesignh, and the rated heat output of a supplementary heater Psup is equal to the supplementary capacity for heating sup(Tj).
(**) If Cdh is not determined by measurement then the default degradation coefficient is Cdh = 0,9.

Technical parameters

Model(s):	Outdoor unit: MHA-V16W/D2N1 Indoor unit: SMK-160/CD30GN1-B
Air-to-water heat pump:	YES
Water-to-water heat pump:	NO
Brine-to-water heat pump:	NO
Low-temperature heat pump:	NO
Equipped with a supplementary heater:	YES
Heat pump combination heater:	NO
Declared climate condition:	COLDER

Parameters are declared for medium-temperature application.

Item	Symbol	Value	Unit
Rated heat output (*)	Prated	16	kW
Declared capacity for heating for part load at indoor temperature 20 °C and outdoor temperature Tj			
Tj = -7℃	Pdh	10.4	kW
Tj = 2℃	Pdh	6.3	kW
Tj = 7℃	Pdh	4.0	kW
Tj = 12℃	Pdh	4.0	kW
Tj = bivalent temperature	Pdh	11.6	kW
Tj = operating limit	Pdh	7.3	kW
For air-to-water heat pumps: Tj = -15℃	Pdh	-	kW
Bivalent temperature	T_{biv}	-12	°C
Cycling interval capacity for heating	P_{cych}	-	kW
Degradation co-efficient (**)	C_{dh}	0.9	--
Power consumption in modes other than active mode			
Off mode	P_{off}	0.019	kW
Standby mode	P_{sb}	0.019	kW
Thermostat-off mode	P_{to}	0.078	kW
Crankcase heater mode	P_{ck}	0.014	kW

Item	Symbol	Value	Unit
Seasonal space heating energy efficiency	ηs	114	%
Declared coefficient of performance or primary energy ratio for part load at indoor temperature 20 °C and outdoor temperature Tj			
Tj = -7℃	COPd	2.44	-
Tj = 2℃	COPd	3.55	-
Tj = 7℃	COPd	5.40	-
Tj = 12℃	COPd	8.20	-
Tj = bivalent temperature	COPd	1.94	-
Tj = operating limit	COPd	1.24	-
For air-to-water heat pumps: Tj = -15℃	COPd	-	-
For air-to-water heat pumps: Operation limit temperature	TOL	-20	°C
Cycling interval efficiency	COP_{cyc}	-	-
Heating water operating limit temperature	W_{TOL}	60	°C
Supplementary heater			
Rated heat output (**)	Psup	8.4	kW
Type of energy input	Electrical		

Other items			
Capacity control	variable		
Sound power level, indoors/ outdoors	L_{WA}	-	dB
Annual energy consumption	Q_{HE}	13244	kWh

For air-to-water heat pumps: Rated air flow rate, outdoors	–	6500	m³/h
For water- or brine-to-water heat pumps: Rated brine or water flow rate, outdoor heat exchanger	–	-	m^3/h

For heat pump combination heater:							
Declared load profile	-			**Water heating energy efficiency**	η_{wh}	-	%
Daily electricity consumption	Q_{elec}	-	kWh	Daily fuel consumption	Q_{fuel}	-	kWh
Annual electricity consumption	AEC	-	kWh	Annual fuel consumption	AFC	-	GJ

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(**) If Cdh is not determined by measurement then the default degradation coefficient is Cdh = 0,9.

Technical parameters

Model(s):	Outdoor unit: MHA-V16W/D2N1 Indoor unit: SMK-160/CD30GN1-B
Air-to-water heat pump:	YES
Water-to-water heat pump:	NO
Brine-to-water heat pump:	NO
Low-temperature heat pump:	NO
Equipped with a supplementary heater:	YES
Heat pump combination heater:	NO
Declared climate condition:	WARMER

Parameters are declared for medium-temperature application.

Item	Symbol	Value	Unit
Rated heat output (*)	Prated	15	kW
Declared capacity for heating for part load at indoor temperature 20 °C and outdoor temperature Tj			
Tj = -7℃	Pdh	-	kW
Tj = 2℃	Pdh	15.3	kW
Tj = 7℃	Pdh	9.9	kW
Tj = 12℃	Pdh	4.4	kW
Tj = bivalent temperature	Pdh	15.3	kW
Tj = operating limit	Pdh	15.3	kW
For air-to-water heat pumps: Tj = -15℃	Pdh	-	kW
Bivalent temperature	T_{biv}	2	°C
Cycling interval capacity for heating	P_{cych}	-	kW
Degradation co-efficient (**)	C_{dh}	0.9	--
Power consumption in modes other than active mode			
Off mode	P_{off}	0.019	kW
Standby mode	P_{sb}	0.019	kW
Thermostat-off mode	P_{to}	0.078	kW
Crankcase heater mode	P_{ck}	0.014	kW

Item	Symbol	Value	Unit
Seasonal space heating energy efficiency	ηs	179	%
Declared coefficient of performance or primary energy ratio for part load at indoor temperature 20 °C and outdoor temperature Tj			
Tj = -7℃	COPd	-	-
Tj = 2℃	COPd	2.34	-
Tj = 7℃	COPd	3.81	-
Tj = 12℃	COPd	6.16	-
Tj = bivalent temperature	COPd	2.34	-
Tj = operating limit	COPd	2.34	-
For air-to-water heat pumps: Tj = -15℃	COPd	-	-
For air-to-water heat pumps: Operation limit temperature	TOL	2	°C
Cycling interval efficiency	COP_{cyc}	-	-
Heating water operating limit temperature	W_{TOL}	60	°C
Supplementary heater			
Rated heat output (**)	Psup	0	kW
Type of energy input	Electrical		

Other items			
Capacity control	variable		
Sound power level, indoors/ outdoors	L_{WA}	-	dB
Annual energy consumption	Q_{HE}	4483	kWh

For air-to-water heat pumps: Rated air flow rate, outdoors	–	6500	m^3/h
For water- or brine-to-water heat pumps: Rated brine or water flow rate, outdoor heat exchanger	–	-	m^3/h

For heat pump combination heater:							
Declared load profile	-			**Water heating energy efficiency**	η_{wh}	-	%
Daily electricity consumption	Q_{elec}	-	kWh	Daily fuel consumption	Q_{fuel}	-	kWh
Annual electricity consumption	AEC	-	kWh	Annual fuel consumption	AFC	-	GJ

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(**) If Cdh is not determined by measurement then the default degradation coefficient is Cdh = 0,9.